

Netflix Movies and TV Shows Data Analysis

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Challenge: Hi! PARIS Machine Learning Research Engineer Challenge

Code repository: <https://github.com/MrLIChuan/hiparischallenge2026>

1. Objective

The objective of this challenge is to explore a Netflix-related movie and TV-show catalogue using data analysis, visualization, and simple machine learning techniques. The analysis focuses on understanding the structure of the catalogue, identifying dominant genres and languages, studying popularity and rating patterns, and building two lightweight baselines: a content-based recommendation system and a popularity prediction model.

The provided dataset contains 9,837 records and 10 columns: `release_date`, `title`, `overview`, `popularity`, `vote_count`, `vote_average`, `original_language`, `genre`, `poster_url`, and `release_year`. The dataset includes rich textual and categorical metadata, as well as numerical indicators related to popularity and user votes. However, it does not contain an explicit type variable distinguishing movies from TV shows, which limits direct comparison between content formats.

2. Data Loading and Preprocessing

The original CSV file presented parsing issues due to malformed rows. To ensure reproducibility, I implemented a robust loading procedure using a Python-based CSV parser, with explicit handling of encoding errors and problematic lines. After loading, the dataset was standardized by normalizing column names, parsing the `release_date` field into a proper datetime format, and extracting `release_year`.

The dataset is relatively complete. Missing values are limited: 10 missing values for `release_date`, `popularity`, and `release_year`; 11 missing values for `vote_count`, `vote_average`, and `poster_url`. Textual fields such as `title`, `overview`, `original_language`, and `genre` are complete. Duplicate handling and basic type checks were also performed to ensure consistency.

Since the `genre` field contains multi-label categories, I transformed it into an exploded genre-level table. This makes it possible to analyze each genre independently, instead of treating multi-genre combinations as unique categories.

3. Exploratory Data Analysis

The catalogue is highly dominated by English-language content. English represents 7,569 titles, followed by Japanese with 645 titles, Spanish with 339 titles, French with 292 titles, and Korean with 170 titles. This distribution suggests a strong concentration around English-speaking productions, while still reflecting some international diversity.

Genre analysis shows that the most frequent genres are Drama, Comedy, Action, Thriller, and Adventure. After splitting multi-label genres, the top categories are:

Drama: 3,744 titles

Comedy: 3,031 titles

Action: 2,686 titles

Thriller: 2,488 titles

Adventure: 1,853 titles

Romance: 1,476 titles

Horror: 1,470 titles

Animation: 1,438 titles

Family: 1,414 titles

Fantasy: 1,308 titles

This suggests that the catalogue is broad but centered around mainstream entertainment categories. Drama is the largest genre, while Comedy, Action and Thriller also represent a significant share of the content.

The numerical indicators reveal a skewed distribution of popularity. The average popularity score is 40.32, but the median is only 21.19, while the maximum reaches 5,083.95. This indicates a long-tail effect: most titles have moderate popularity, while a very small number of titles are extremely popular. A similar concentration is observed in `vote_count`, where the average is 1,392.94 but the maximum reaches 31,077. The average rating is more stable. The mean `vote_average` is 6.44, with a standard deviation of 1.13. Most titles are therefore rated around the middle-to-high range, while very low and very high ratings are less common. When comparing genres by average popularity, Adventure, Fantasy, Animation, Action, and Science Fiction obtain the highest average popularity scores. In contrast, when comparing genres by average rating, History, War, Music, Animation, and Western rank highest. This difference is interesting: the most popular genres are not necessarily the highest-rated ones. Popularity appears to be more associated with mainstream and visually engaging genres, while ratings may reward more niche or critically appreciated genres.

4. Recommendation and Machine Learning Baselines

I implemented a simple content-based recommendation system using TF-IDF text representations. The text representation combines `title`, `genre`, `overview`, and `original_language`. The resulting TF-IDF matrix has shape $9,837 \times 10,000$. Cosine similarity is then used to retrieve the most similar titles for a given input title.

This method is simple, interpretable, and reproducible. It does not require user interaction data, which makes it suitable for cold-start scenarios. However, it only captures textual similarity and does not model user preferences or viewing behavior.

I also implemented a baseline regression model to predict popularity. The model uses `release_year`, `vote_average`, `vote_count`, `original_language`, and `genre` as features. A Random Forest Regressor was trained with preprocessing for numerical and categorical variables. The resulting performance was: Mean Absolute Error: 23.2183 ; R^2 score: 0.2318

The R^2 score indicates that the available metadata explains only part of the variance in popularity. This is expected because popularity is likely influenced by external factors not included in the dataset, such as marketing campaigns, platform exposure, cast, franchise effects, social media trends, and release timing.

5. Key Insights and Limitations

The analysis highlights several findings. First, the catalogue is strongly dominated by English-language content. Second, Drama, Comedy, Action and Thriller are the most represented genres. Third, popularity is highly skewed, with a small number of titles concentrating very high attention. Fourth, genre-level analysis shows a distinction between popular genres and highly rated genres. There are also several limitations. The dataset does not provide a clear distinction between movies and TV shows, despite the challenge title. It also lacks user-level interaction data, which prevents collaborative filtering or personalized recommendation. In addition, popularity is difficult to model accurately using only catalogue metadata. Finally, the dataset appears to contain movie-database-style metadata rather than detailed Netflix platform-level information.

6. Reproducibility

The full analysis is provided as a documented and reproducible GitHub repository. The code includes data loading, preprocessing, exploratory data analysis, visualizations, a TF-IDF recommendation baseline, and a Random Forest popularity prediction baseline. Dependencies are listed in `requirements.txt`, and the notebook can be executed end-to-end.